

ComplaintOps

From customer voice to operating decisions

A reproducible analysis of confidence-aware complaint routing, queue forecasting, workforce capacity, and customer support design.

6,165

synthetic complaints

81.8%

routing accuracy

9.3%

average forecast WAPE

9

base-plan agents

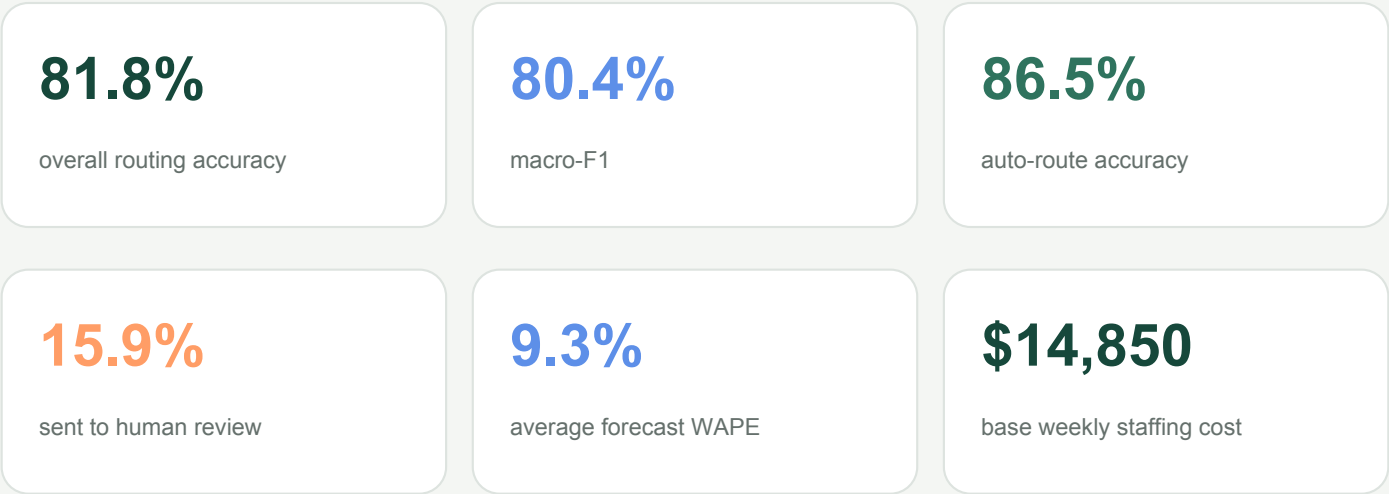
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What the project demonstrates

ComplaintOps connects three decisions that are often separated in portfolio work: where a complaint should go, how much work each queue should expect, and how much capacity should be scheduled. A customer-facing prototype adds explanation, human review, and feedback without pretending the demo is a live service.



CORE BUSINESS INTERPRETATION

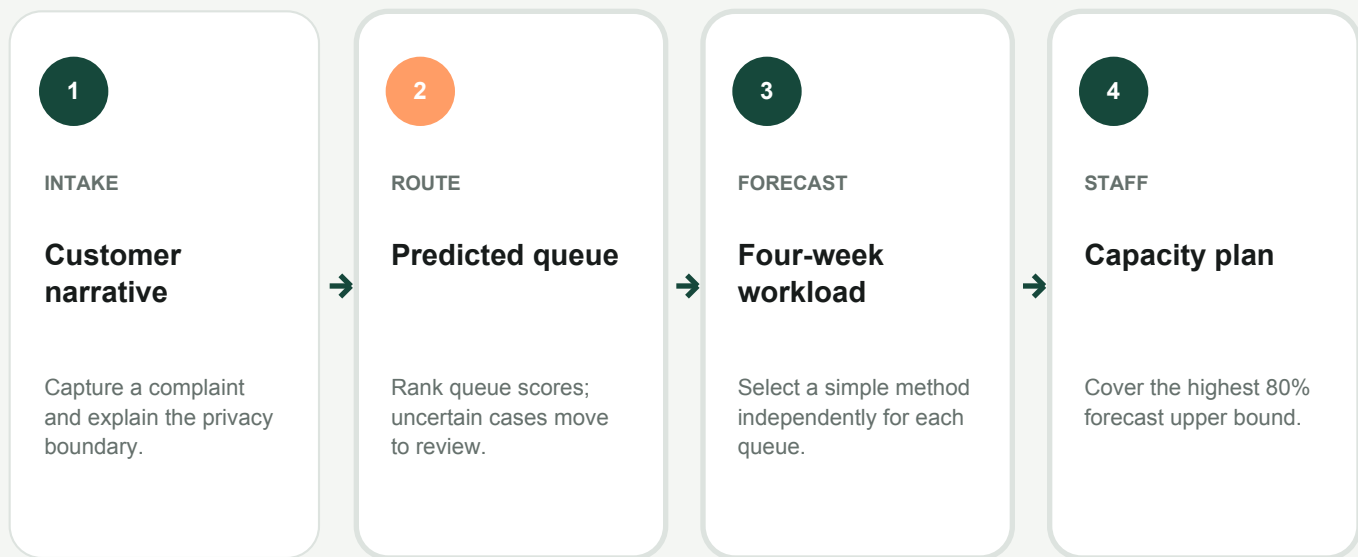
The model should not force every case into automation. A calibration-selected threshold routes 84.1% of final test cases automatically at 86.5% accuracy and reserves the remaining 15.9% for review. Forecast uncertainty then flows into an explicit nine-agent capacity plan.

Scope boundary

Every metric in this report comes from a deterministic synthetic dataset. The purpose is to demonstrate analytical reasoning, validation, and product design - not to claim production performance or customer impact.

One analysis, four connected decisions

The project is structured as a decision pipeline. Each output becomes an input to the next stage, so the analysis ends with an operating recommendation rather than a collection of unrelated charts.



CUSTOMER EXPERIENCE LAYER

The interactive application lets a user submit a sample narrative, view ranked routing scores, request a human specialist, open a device-local demo case, ask guided support questions, and leave a 1-5 rating with written feedback. No real complaint text is transmitted.

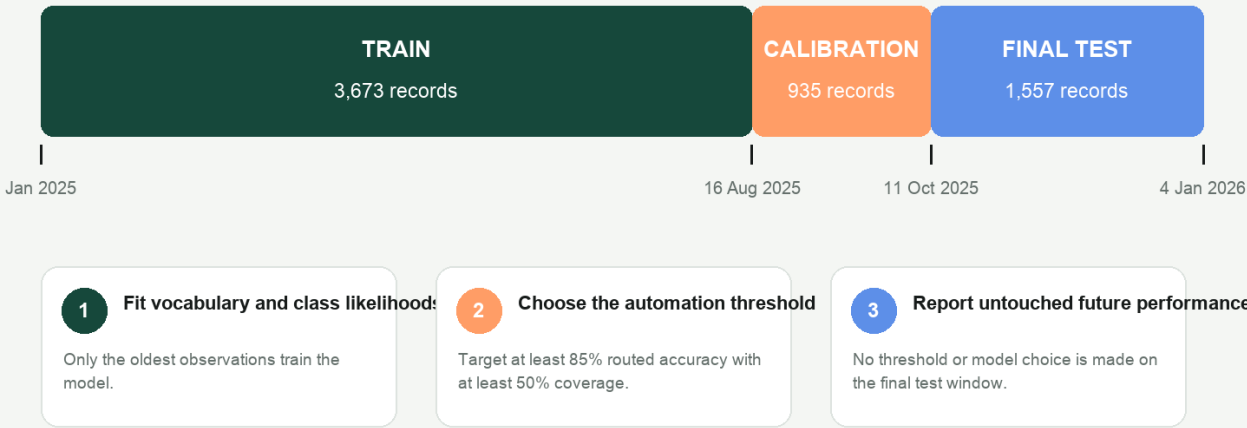
Why this matters for business analytics

The same model can be technically accurate yet operationally poor if it hides uncertainty, provides no escalation path, or turns forecast averages into false staffing precision. ComplaintOps makes those policy choices visible.

Temporal validation without test leakage

Chronological validation design

The threshold is chosen on calibration data and frozen before the newest test period.



ComplaintOps | Synthetic demonstration data

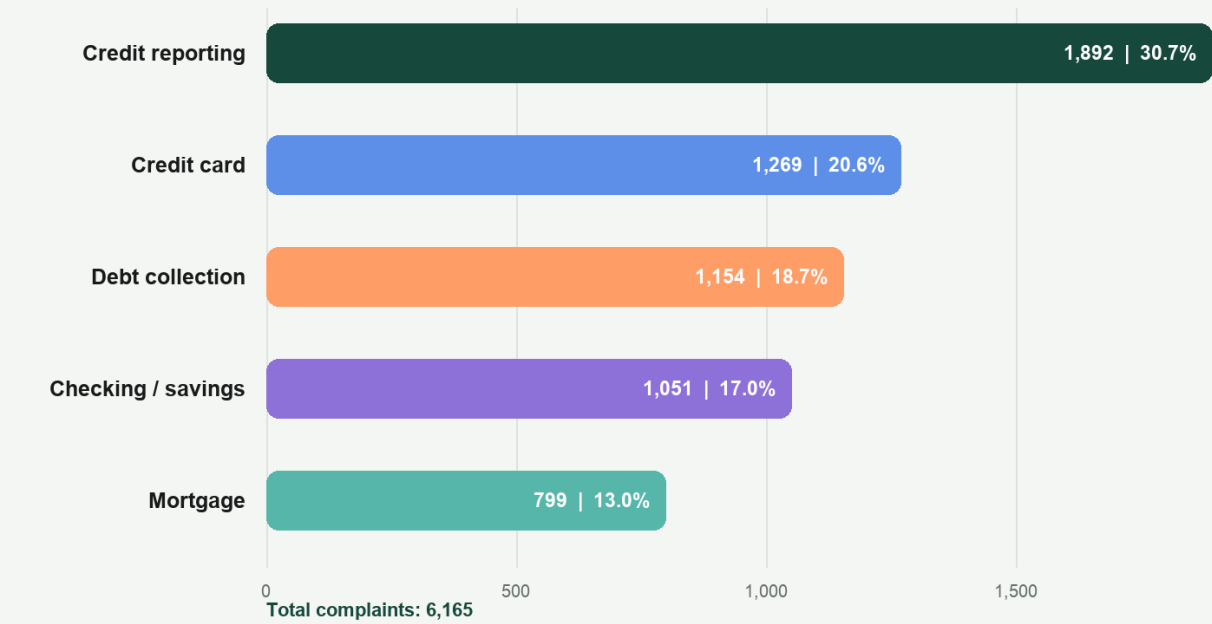
Validation rule

The model learns from 3,673 older cases. A separate 935-case calibration window chooses the 0.9617 automation threshold. Only then is performance reported on 1,557 newer cases.

Where the workload is concentrated

Complaint portfolio mix

Volume concentration determines where routing errors and staffing gaps matter most.



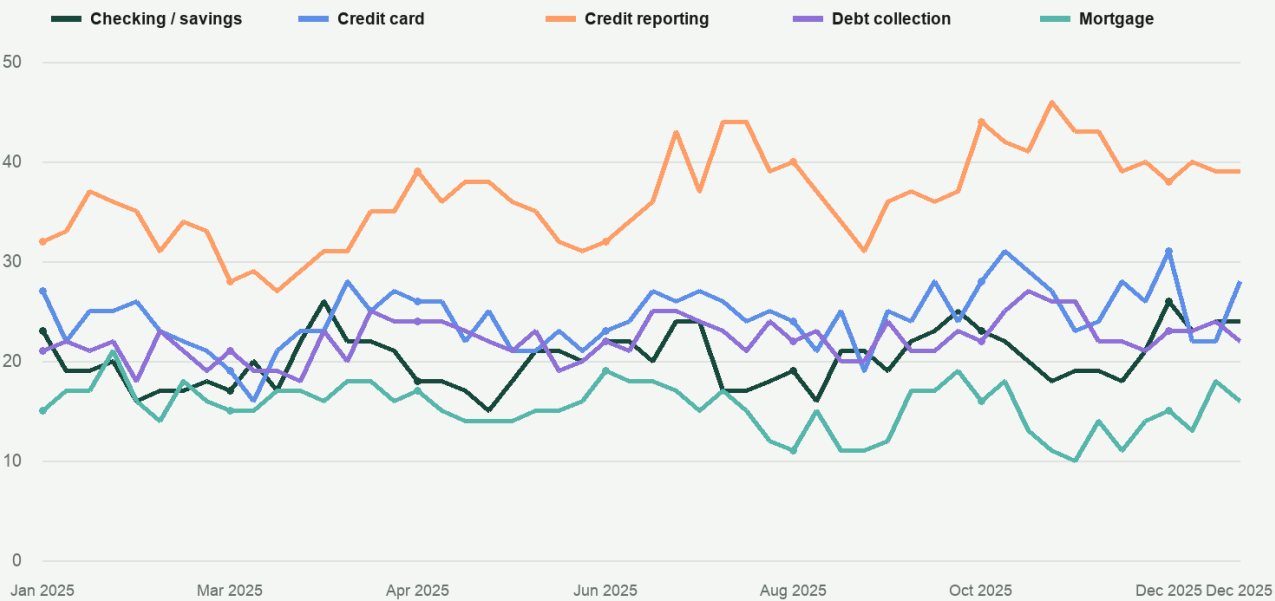
Management readout

Credit reporting contributes 1,892 cases, or 30.7% of the portfolio. Its scale makes even a small routing or forecast error operationally material; mortgage is smaller but still requires dedicated coverage under the current assumptions.

How weekly queue demand changes

Weekly complaint demand

Fifty-two weeks of synthetic volume reveal queue-specific scale and variability.



ComplaintOps | Synthetic demonstration data

Planning readout

The five queues move at different levels and with different short-run patterns. Forecasting each queue independently is more defensible than applying one aggregate growth assumption to the entire operation.

Where the classifier makes mistakes

Routing confusion matrix

Rows are actual queues; cells show row share with record count on the final temporal test set.

		Checking / savings	Credit card	Credit reporting	Debt collection	Mortgage
Actual queue	Checking / savings	83% n=221	3% n=8	5% n=12	6% n=15	4% n=10
	Credit card	7% n=21	82% n=262	3% n=10	6% n=19	3% n=9
	Credit reporting	2% n=11	4% n=18	85% n=432	6% n=31	3% n=15
	Debt collection	6% n=18	5% n=15	4% n=12	78% n=227	6% n=18
	Mortgage	7% n=12	5% n=9	8% n=14	3% n=6	76% n=132
		Predicted queue				

ComplaintOps | Synthetic demonstration data

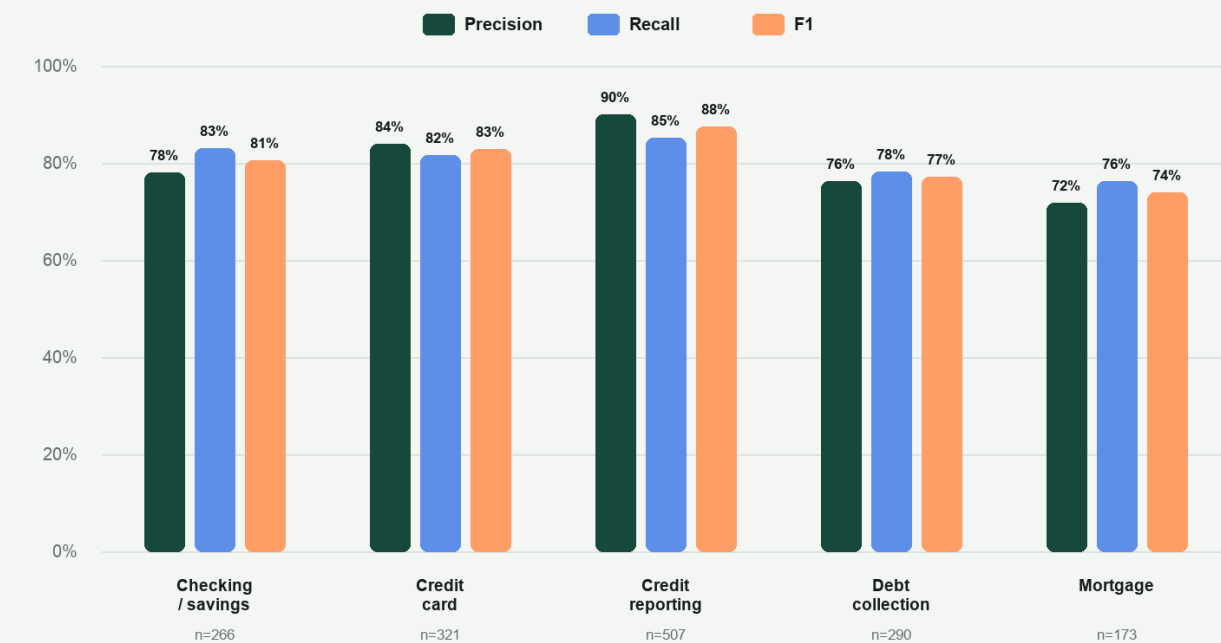
Error pattern

Credit reporting has the strongest test recall at 85%. Mortgage is hardest at 76%, with the largest single error share moving to credit reporting. These off-diagonal cells identify where human-review guidance and targeted data collection should focus.

Balanced performance across queues

Routing quality by product queue

Macro-F1 prevents the largest class from hiding weaker performance in smaller queues.



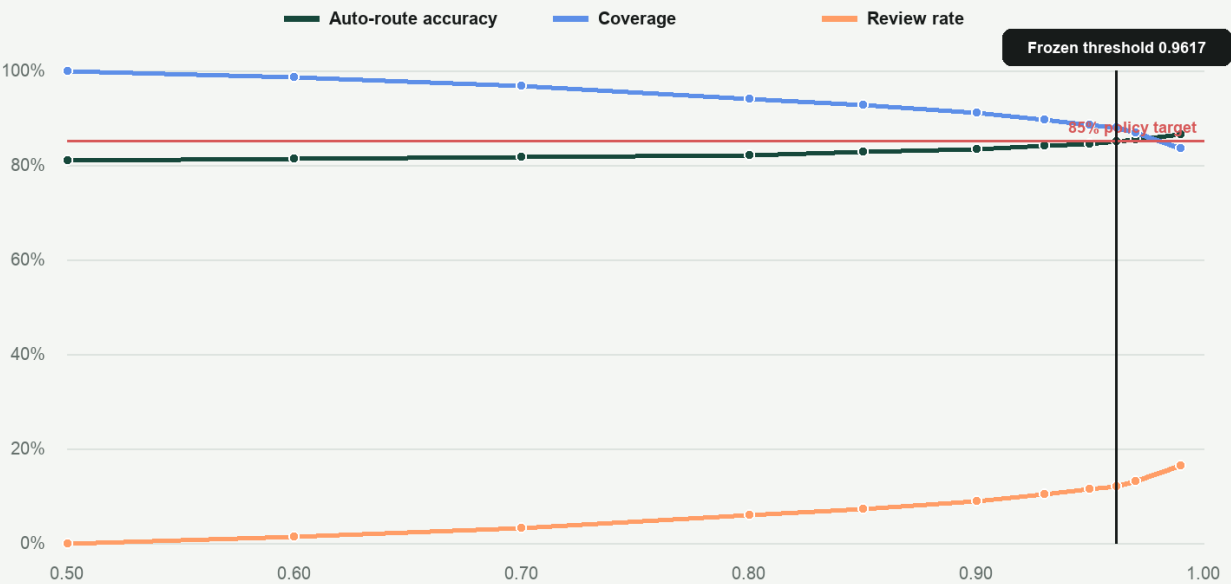
Why macro-F1 is reported

Overall accuracy can be dominated by the largest class. The 80.4% macro-F1 gives equal weight to all five queues and exposes the lower mortgage precision and F1 that a single accuracy figure would hide.

Trading automation coverage for quality

Confidence threshold trade-off

A higher threshold improves automated-route accuracy but sends more cases to human review.



ComplaintOps | Synthetic demonstration data

Frozen policy result

The calibration window selected 0.9617 as the highest-coverage threshold meeting the 85% routed-accuracy target. On the untouched test period, that policy auto-routes 84.1% of cases at 86.5% accuracy and reviews 15.9%.

Selecting a baseline by queue

Forecast model comparison

Each queue selects the lowest walk-forward WAPE; complexity is used only when it improves the baseline.

	NAIVE	MA4	MA8	TREND8
Checking / savings	9.8% SELECTED	11.8%	13.9%	13.2%
Credit card	11.2%	11.0% SELECTED	11.6%	12.8%
Credit reporting	6.7% SELECTED	8.1%	10.2%	10.0%
Debt collection	7.6% SELECTED	8.0%	8.5%	8.5%
Mortgage	11.7% SELECTED	12.9%	14.8%	15.6%

Metric: weighted absolute percentage error (lower is better)

ComplaintOps | Synthetic demonstration data

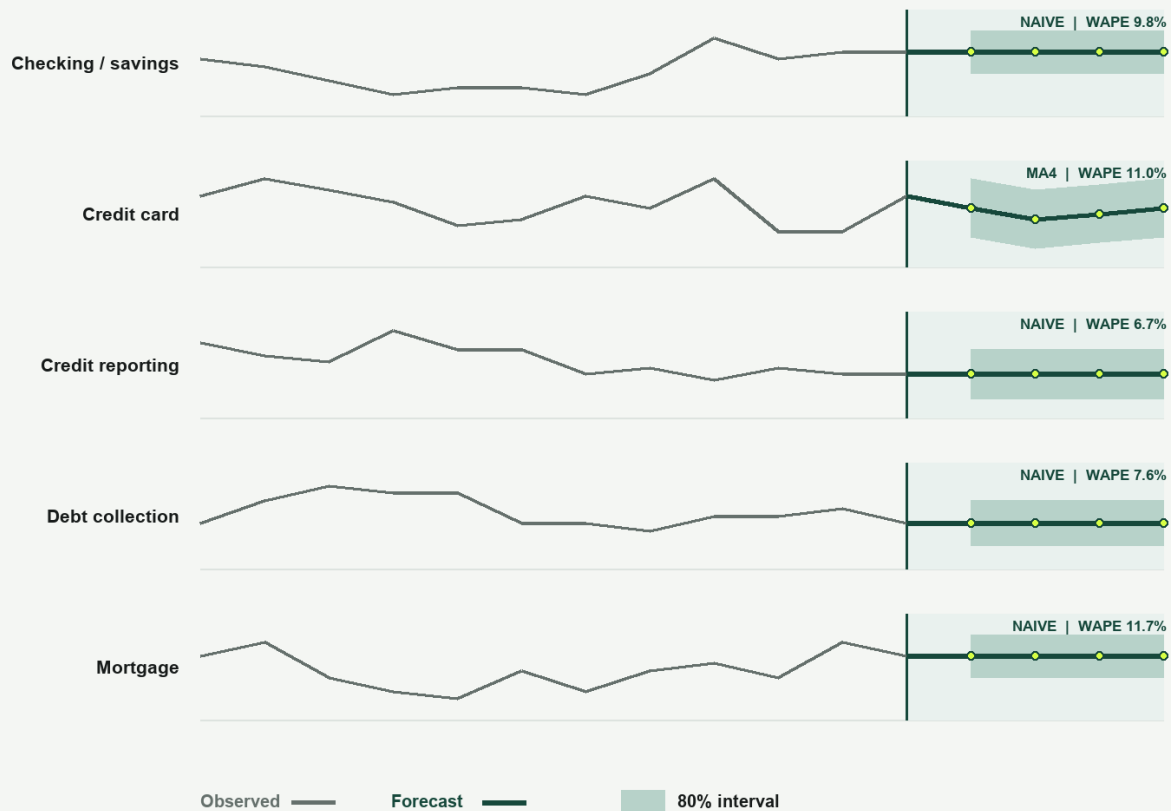
Model-selection result

The naive forecast wins four queues. Only credit card benefits from the four-week moving average, and the improvement is modest. This is evidence against adding complexity merely to make the project look more advanced.

Four-week operating outlook

Four-week demand outlook

The selected forecast is shown with an empirical 80% error interval for capacity planning.



ComplaintOps | Synthetic demonstration data

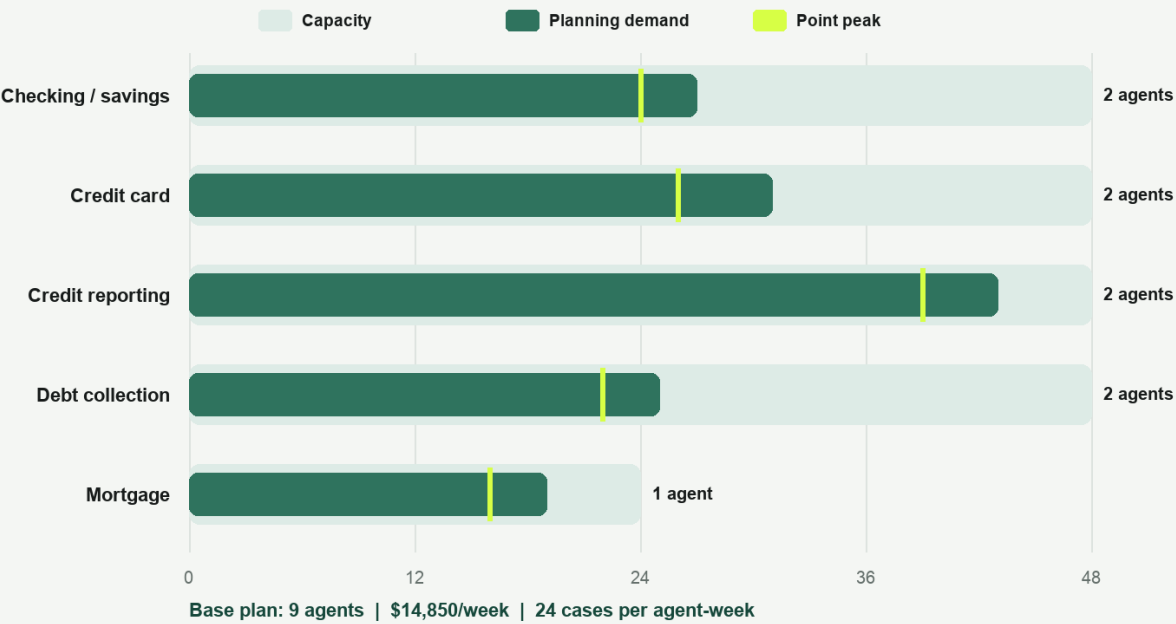
Decision use

Point forecasts support daily visibility, but the capacity decision uses the empirical 80% upper bound. Credit reporting has the highest four-week point forecast at 39 cases per week and an upper bound of 43.

Capacity against uncertainty

Workforce capacity by queue

Recommended headcount covers the peak 80% upper bound under dedicated-team assumptions.



ComplaintOps | Synthetic demonstration data

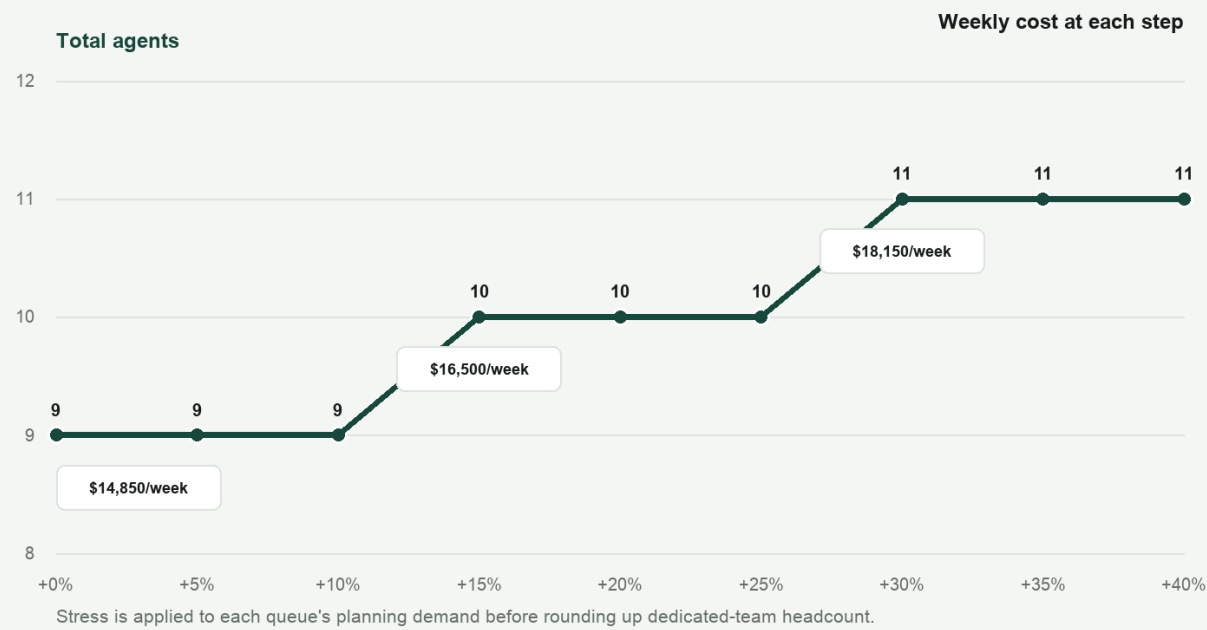
Base-plan recommendation

At 24 cases per agent-week and \$1,650 per agent-week, the dedicated-team scenario requires nine agents and \$14,850 weekly. Visible slack is a consequence of integer headcount and queue separation, not hidden efficiency.

When volume stress changes headcount

Demand-stress sensitivity

Discrete headcount steps show where a modest volume shock creates a real capacity decision.



ComplaintOps | Synthetic demonstration data

Sensitivity result

The plan stays at nine agents through a 10% demand stress, rises to ten at 15%, and rises to eleven at 30%. The step pattern is important: small forecast changes do not always imply a real staffing action.

Recommendations and reproducibility

DECISION RECOMMENDATIONS

- 1 Keep the confidence-aware review path; do not optimize only for raw automation coverage.
- 2 Benchmark every queue against the naive forecast before adopting a more complex method.
- 3 Treat the nine-agent result as a transparent scenario, not a fixed staffing promise.
- 4 Monitor class drift, review rate, forecast error, backlog age, and threshold stability together.

LIMITATIONS TO DEFEND IN AN INTERVIEW

- Synthetic data cannot establish external validity or real customer impact.
- Naive Bayes confidence is a routing score, not a perfectly calibrated probability.
- Forecast WAPE is used for model selection and is not a separate untouched forecast holdout.
- Dedicated teams, fixed productivity, and no backlog or schedule constraints simplify workforce planning.
- The support assistant is rule-based and device-local; it is not a live customer-service channel.

REPRODUCE THE SUBMISSION

```
$ make check  
$ python -m pip install -r requirements-viz.txt  
$ make visualizations  
$ make report
```

Repository: github.com/Simon-Yu-analytics/Complaintops

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